Analyzing and Predicting Song Popularity Using Spotify Audio Features

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***Abstract*— This interim report explores song popularity in the SpotGenTrack dataset from two complementary angles. First, it examines whether popularity differs across songs that display different levels of energy and danceability. Second, it investigates how well popularity can be estimated from a selected group of Spotify audio features within a regression framework. To support these aims, the study focuses on the Spotify track-level table and performs a structured profiling process including feature selection, descriptive statistics, duplicate checks, missing-value inspection, correlation analysis, skewness assessment, and outlier exploration. In addition, a simple mean-based baseline model is established for the predictive task, and suitable regression metrics are defined. The report provides the analytical foundation for the final phase of the project, where formal hypothesis testing and machine learning models will be implemented.**

Keywords— song popularity, Spotify audio features, regression, hypothesis testing, data profiling, exploratory analysis

I. Introduction

The expansion of music streaming platforms has made it possible to study musical success through large-scale digital data. Spotify, in particular, provides track-level features such as danceability, energy, speechiness, tempo, and valence, which make it possible to examine whether measurable audio characteristics are associated with song popularity. In this study, the SpotGenTrack dataset is used to analyze popularity from two related perspectives: differences across audio profiles and predictive modeling based on selected Spotify features.

Prior studies support this direction. Kim and Oh found that granular Spotify-style acoustic variables improve the prediction of top-ten-hit songs, indicating that detailed audio features carry substantial predictive value [1]. This is especially relevant to the present study because it supports the use of track-level Spotify variables and provides a foundation for examining whether popularity varies across songs with different energy and danceability profiles. More recent work by Zeng, Yuan, Guo, and Xu also showed that popularity prediction can benefit from carefully selected feature representations [2]. Although their study used a broader multimodal framework, it still supports the idea that a structured set of predictors can be effective for modeling popularity. In a related study, Oliveira, Couto da Silva, and Moro found that audio-related characteristics are associated with popularity-related outcomes in digital music environments [3]. Although their focus was music virality rather than Spotify popularity alone, their findings support the broader assumption that measurable musical properties help explain success on streaming platforms.

Taken together, these studies suggest two main ideas. First, Spotify-style audio features are meaningfully related to music success and popularity-related outcomes. Second, popularity prediction can be approached effectively through a selected set of structured audio variables. Building on this literature, the present study addresses the following research questions:

RQ1: How does song popularity vary across tracks with different combinations of energy and danceability?

RQ2: How accurately can song popularity be predicted using danceability, energy, speechiness, tempo, valence, and duration_ms?

These questions are closely connected because both focus on song popularity as the central outcome variable. The first question approaches popularity from an explanatory perspective, while the second examines it from a predictive perspective.

II. Data Profiling and Descriptive Statistics

A. Selected Features

The variables in this study were chosen in line with the two research questions.

For the first question, the main variables are popularity, energy, and danceability. Popularity is the dependent variable, while energy and danceability are used to describe contrasting audio profiles. These two features were selected because they are interpretable, widely used in Spotify-based analyses, and suitable for defining groups in the statistical part of the project.

For the second question, the predictive feature set consists of danceability, energy, speechiness, tempo, valence, and duration_ms. These variables reflect different aspects of musical structure. Danceability and tempo relate to rhythmic character, energy reflects intensity, speechiness captures the degree of spoken-word content, valence represents emotional tone, and duration_ms reflects track length. Together, they provide a compact and manageable predictor set for the regression task.

The variables were selected primarily on the basis of the research design and the literature. In addition, correlation analysis was used during the profiling stage to check whether the chosen predictors showed meaningful relationships with popularity and whether any strong redundancy was present among them.

B. Descriptive Statistics

Descriptive statistics were examined before moving to formal analysis. This step was useful for understanding the scale, spread, and general distribution of the selected variables.

The results indicate that the mean popularity score in the dataset is 39.78, with a standard deviation of 16.79, which suggests considerable variation across tracks. Both energy and danceability have average values close to 0.586, and their medians are slightly above the mean, indicating that many songs in the dataset fall in the medium-to-high range on these dimensions.

Among the predictors selected for the regression task, speechiness has a comparatively low average value, while tempo and duration_ms naturally span much wider numerical ranges than the bounded audio indicators. Valence lies around the middle of its range on average, which suggests that the songs in the dataset cover a broad emotional spectrum. Overall, the descriptive patterns indicate that the selected predictors contain enough variability to support predictive modeling.

TABLE I. DESCRIPTIVE STATISTICS OF THE SELECTED VARIABLES

	mean	*std*	*min*	*median*	*max*
popularity	39.782	16.791	0	41	97
energy	0.586	0.26	0	0.629	1
danceability	0.586	0.178	0	0.61	0.989
speechiness	0.129	0.203	0	0.051	0.969
tempo	118.359	30.224	0	118.067	244.035
valence	0.483	0.262	0	0.477	0.993
duration_ms	246770.8	190430.3	1155	216893	5505831

C. *Data Quality Checks*

Several quality checks were carried out before the analysis began. First, the available tables were inspected in terms of structure, variable names, and key columns. The main analytical dataset was built on spotify_tracks.csv, which contains 101,939 observations and a unique identifier for each track. This makes the table suitable as the main source for both parts of the project.

Second, the auxiliary tables were reviewed to determine whether they could be merged safely at the song level. The album table contains a unique track identifier, while the artist table includes repeated track IDs because the same song may be linked to multiple artists. A direct merge with the artist table would therefore duplicate track-level observations. Since the current version of the project relies only on track-level Spotify variables, the auxiliary tables were not included in the interim analysis dataset.

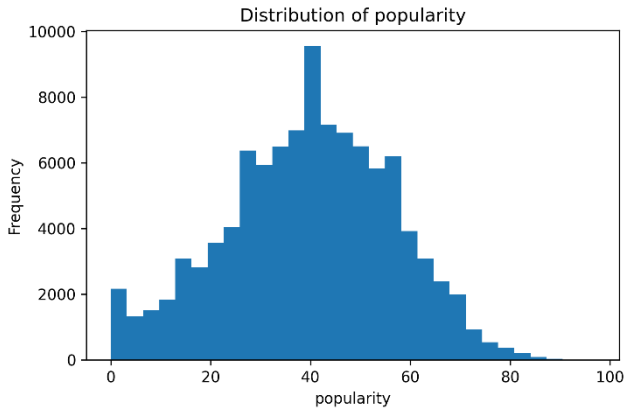
Third, duplicate checks were performed at both the row and key levels. No duplicate rows were found in the main tracks dataset. The selected variables for the present analysis also showed no missing values, which means that the core feature set is complete and ready for use in both the statistical and machine learning stages.

D. *Correlation, Skewness, and Outlier Assessment*

Beyond basic descriptive statistics, additional profiling steps were used to better understand the selected variables.

Pairwise correlations showed that popularity has positive but not very strong relationships with several of the selected Spotify features. This suggests that the chosen variables are relevant, while also indicating that popularity is likely shaped by multiple influences rather than a single dominant characteristic. Moderate correlations were also observed among some predictors, which is not unusual in audio-feature datasets and should be kept in mind when interpreting future models.

Distributional checks were conducted using histograms and boxplots for the main variables in RQ1. As shown in Fig. 1, popularity is concentrated around medium values, with fewer observations at the very high end of the scale. This pattern suggests that song popularity is not evenly distributed across tracks and supports the use of error-based metrics in the predictive stage.



In addition, skewness values and IQR-based outlier summaries were calculated for the selected variables. Some variables showed non-negligible skewness, and a number of observations fell outside the conventional IQR limits. These observations were not removed at this stage because they may reflect real variation in the music data rather than simple recording errors. Their influence will be reconsidered during the final modeling phase if necessary.

III. PERFORMANCE METRICS AND NAIVE BASELINE MODEL

A. *Selected Evaluation Metrics*

The target variable, popularity, is continuous, so model evaluation must be based on prediction error.

For this reason, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were selected as the main performance metrics, while R^2 was included as a complementary measure.

MAE represents the average absolute difference between observed and predicted popularity values. Its main advantage is interpretability, since it stays on the same scale as the target variable. In this project, MAE can be read directly as the average number of popularity points by which the predictions differ from the true values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

RMSE also measures prediction error, but it gives greater weight to larger mistakes because the errors are squared before averaging. This makes RMSE useful when larger misses should be penalized more strongly.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

R^2 measures how much better a model performs than a simple mean-based predictor. A value close to 1 indicates strong fit, a value around 0 indicates performance similar to predicting the mean, and a negative value indicates performance worse than that benchmark.

Taken together, these three metrics offer a balanced view of model performance. MAE provides a stable and intuitive

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (3)$$

measure of average error, RMSE highlights the presence of larger deviations, and R^2 helps evaluate whether a model improves meaningfully over a trivial reference point.

B. Naive Baseline Model

A naive baseline model was defined by predicting the mean popularity value from the training set for every observation in the test set. This approach does not use any explanatory variables and therefore serves as a simple reference for the predictive task.

The main advantage of this baseline is that it is transparent and easy to interpret. It establishes a minimum standard that more advanced models should outperform. If a trained regression model cannot do better than this mean-based rule, then its practical value would be limited.

At the same time, this baseline has an obvious weakness: it assigns the same predicted popularity to every song. As a result, it cannot capture any of the variation associated with differences in audio characteristics. For this reason, it should be viewed only as a lower-bound benchmark.

C. Baseline Results

The naïve baseline was evaluated on the full track-level dataset containing 101,939 songs. After the train-test split, 81,551 observations were used for training and 20,388 observations were used for testing. The mean popularity in the training set was 39.771, and this value was used as the prediction for every song in the test set.

The baseline achieved an MAE of 13.617 and an RMSE of 16.900. This means that a prediction strategy that ignores all features and always returns the training-set mean makes an error of about 13.6 popularity points on average, while the larger-error-sensitive RMSE is about 16.9. The R^2 value is approximately 0, which is expected for a mean-based benchmark because it does not improve meaningfully over the trivial reference used in the definition of R^2.

These results provide a clear minimum benchmark for the prediction problem. In the final phase, trained regression models will be evaluated using the same metrics, and their usefulness will be judged according to whether they can reduce MAE and RMSE and achieve stronger fit relative to this baseline.

TABLE II. BASELINE RESULTS IN THE STUDY

	Baseline Results in the Study
MAE	13.617
RMSE	16.900
R^2	0.000

IV. CONCLUSION

This interim report established the analytical groundwork for the final stage of the project. Two connected research questions were defined around the common theme of song popularity. A focused set of Spotify audio variables was selected, and the data was examined through descriptive statistics, key checks, missing-value inspection, correlation analysis, skewness analysis, and exploratory visualizations.

The interim findings suggest that the selected track-level features are suitable for both explanatory and predictive analysis. In addition, a mean-based naïve regression baseline and appropriate evaluation metrics were established for the machine learning task. The next stage of the project will build on this foundation by conducting formal hypothesis testing for group differences in popularity and by training machine learning regression models to assess how well song popularity can be predicted from selected Spotify audio features.

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